

Agentic Explainable Artificial Intelligence (Agentic XAI) Approach To Explore Better Explanation

AI for earth Observation Course - AI4EO - M2 GeoAI

June, 2026

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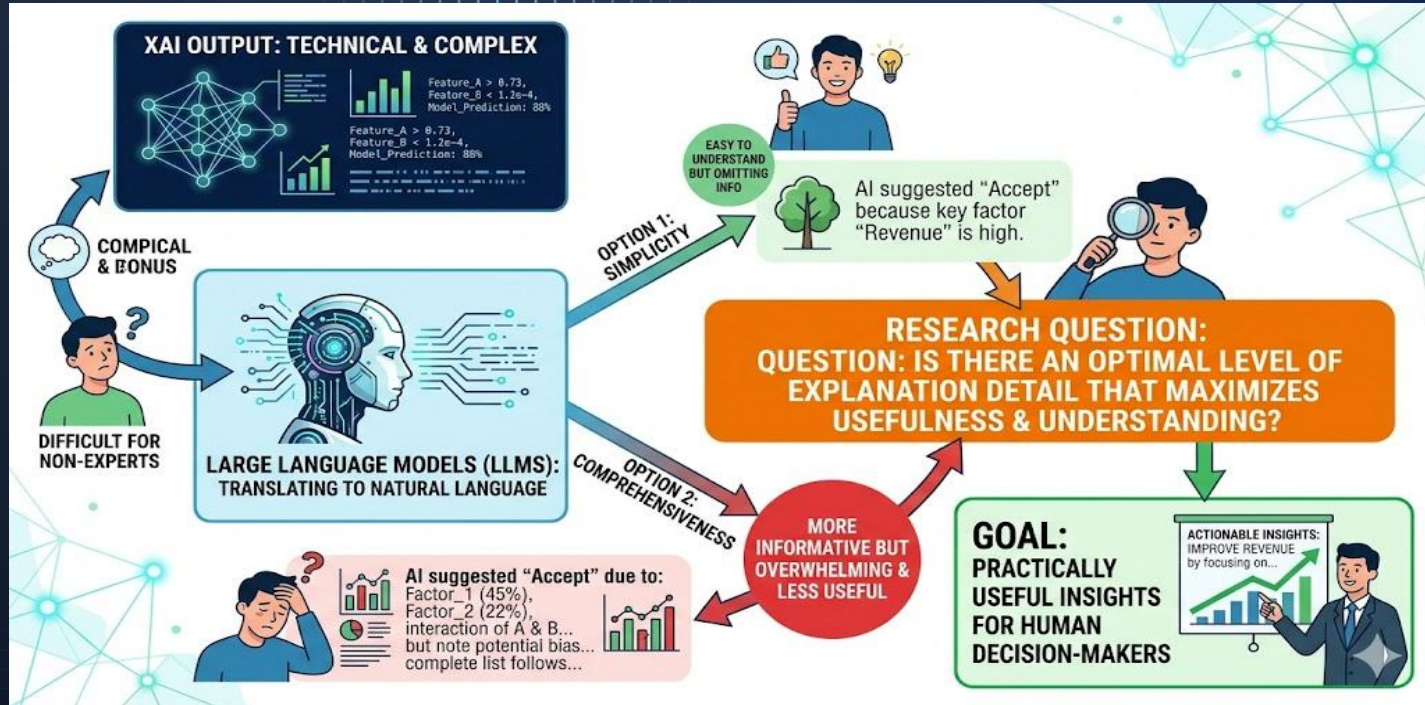
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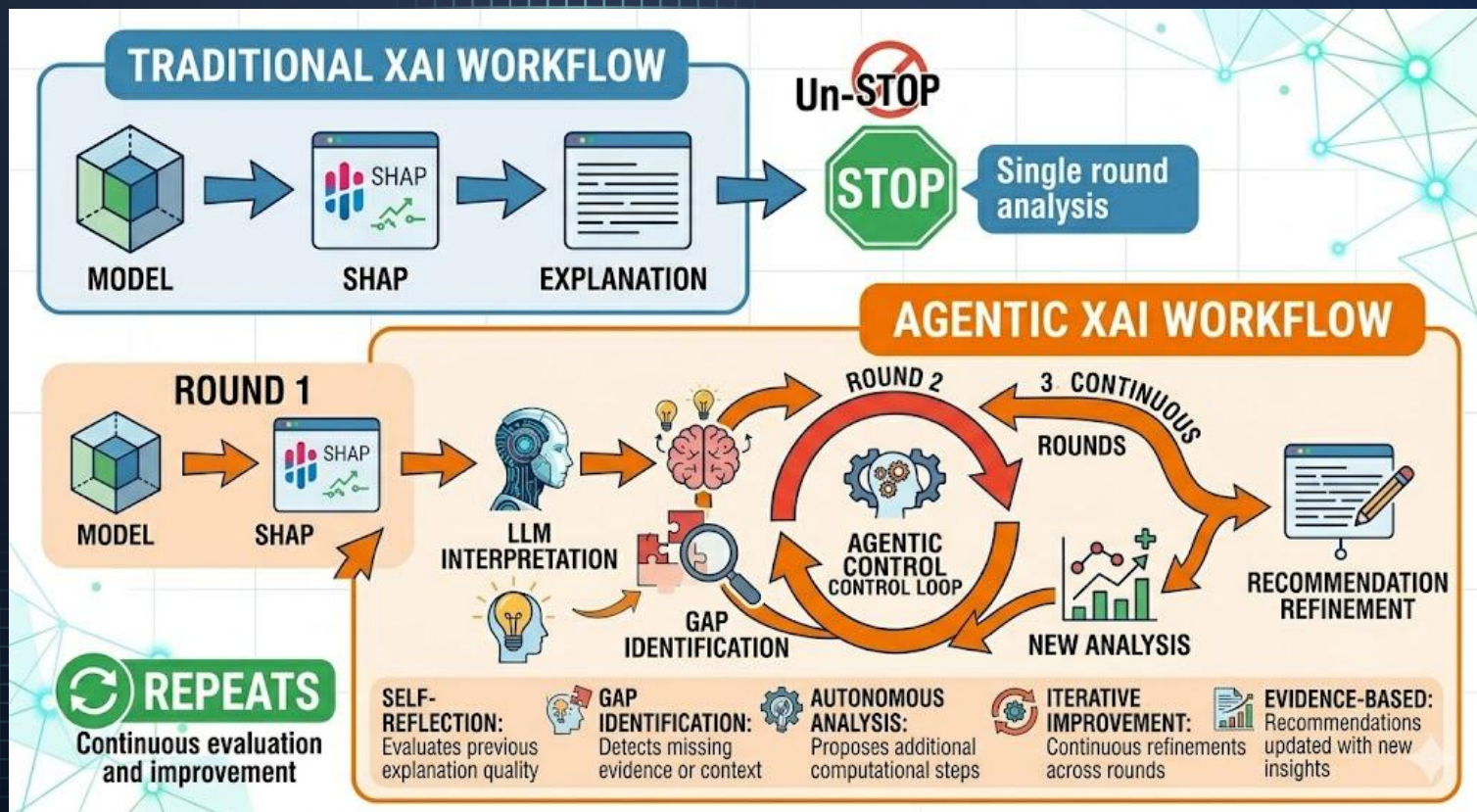
Presented to: Dr Ali Ghandour

Introduction & Motivation



Motivation: Improve the communication of AI-generated insights so that they are both understandable and practically useful for human decision-makers.

Why Is This Considered Agentic AI?



How do we convert technical AI explanations into human-friendly explanations?

Even if we simplify explanations with LLM, How detailed should the explanation be?

Because:

Too little explanation = incomplete

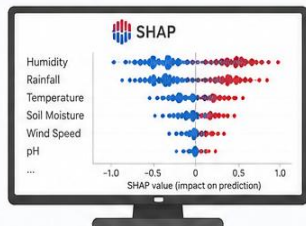
Too much explanation = overwhelming

This is similar to human teaching

The Two-Layered Problem

LAYER 1

The XAI Communication Gap



Technical XAI Output
(e.g., SHAP Beeswarm Plot)



Non-Expert
(Farmer / Practitioner)



Difficult to interpret and act on

Interpretability-Usability Gap

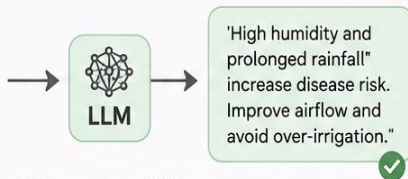


Problem

Technically rigorous XAI outputs are not easily understandable or actionable for non-experts.

LAYER 2

The Unknown Optimal Explanation Depth



LLM translates XAI into plain language

How much explanation is ideal?



Problem

The optimal level of explanation depth is unknown.

No prior work has empirically tested this trade-off.

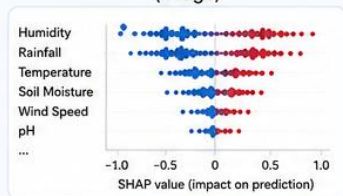
2. The Proposed Methodology: Agentic XAI

Agentic XAI: An AI agent iteratively explores and refines the best possible explanation of modeled patterns through natural language recommendations.

STAGE 1 Initial Interpretation (Round 0)

Inputs

SHAP Beeswarm Plot
(Image)



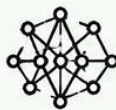
+

Variable List
(Text)

- Humidity
- Rainfall
- Temperature
- Soil Moisture
- Wind Speed
- pH
- ..



Multimodal LLM
(Claude Sonnet 4)



Claude



Output: Baseline Recommendation (Round 0)

"High humidity and prolonged rainfall appear to be the strongest drivers of disease risk. Farmers should monitor irrigation and improve airflow between crops."

(Baseline for further refinement)



XAI Analysis (Understanding the data)

A SHAP beeswarm plot visualization show:

- which features matter most
 - whether they increase/decrease predictions
- distribution of impacts



Anthropic's Claude Sonnet 4.

It is "multimodal MLLM" meaning it can understand: text, images, charts, PDFs
So the LLM looks at the SHAP graph like a human researcher would



The AI produces an initial recommendation. This is the baseline explanation. But the system does NOT stop there.

This is the innovative part.
The paper runs multiple improvement cycles.
Each round tries to improve the explanation.

STAGE 2 Iterative Refinement Loop (Rounds 1–10)

1 Gap Analysis & Code Generation



The LLM reviews the previous recommendation and identifies what is missing or unclear.

“What additional analyses are needed to strengthen this explanation?”



The LLM writes Python code to perform supplementary analyses (e.g., correlations, PCA, clustering, feature interactions).

2 Code Execution & Evidence Generation



The generated code is executed on the real dataset.
Outputs (tables, plots, statistics) are compiled into a PDF report.

Examples of Analyses



Correlations



PCA



Clustering

...

and more



3 Synthesis & Recommendation Refinement



The LLM receives:

- Original SHAP plot (image)
- PDF with new analyses
- Full conversation history

It synthesizes the new evidence and rewrites the recommendations with greater depth and clarity.



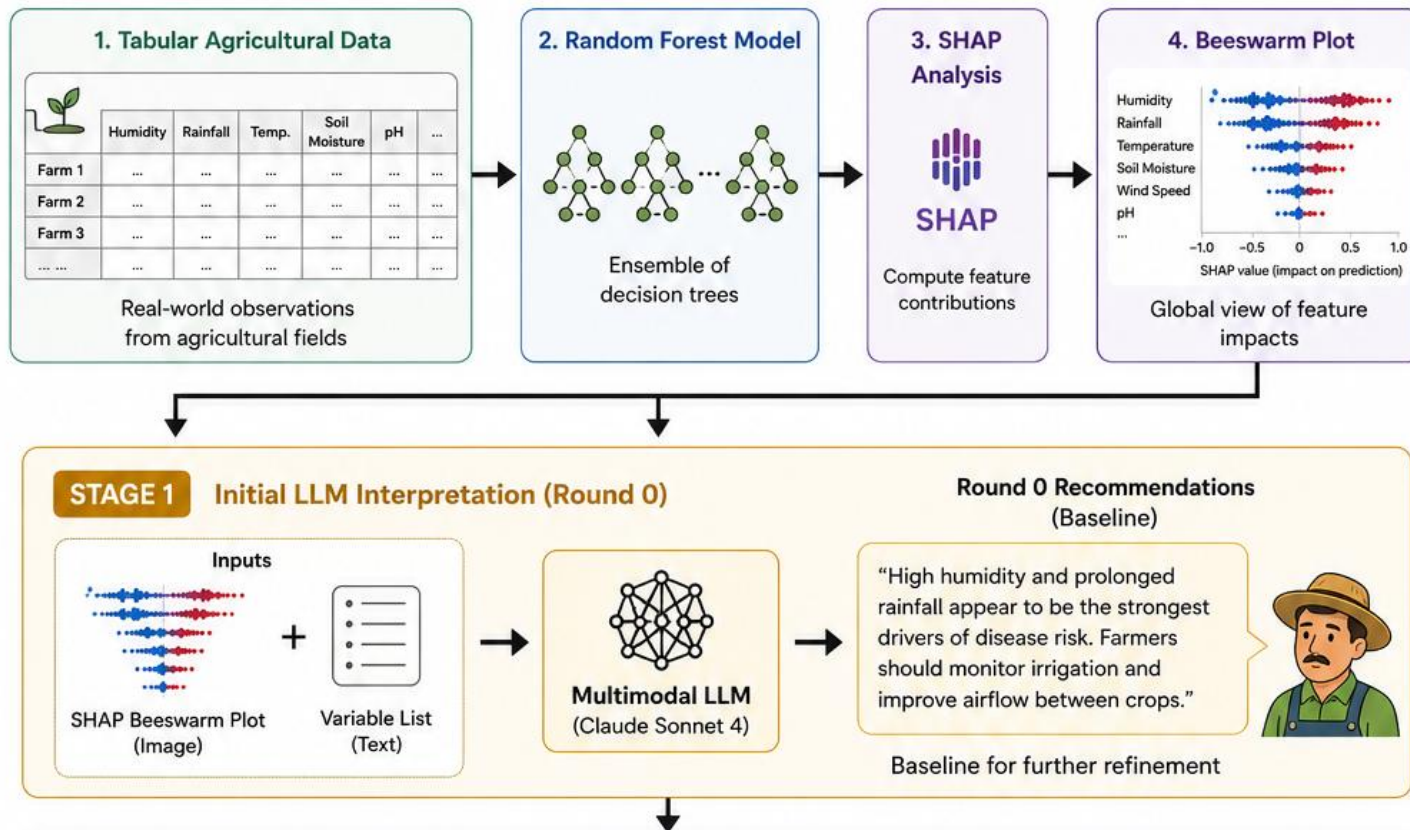
Repeat for Rounds 1 → 10
(each round builds on all previous context and evidence)



Final Output (After Round 10): Most Refined Recommendations for Farmers
More comprehensive, evidence-based, and actionable.



3. OVERALL SYSTEM ARCHITECTURE



Random Forest handles this well because it:

- captures nonlinear relationships
- is robust to noise
- works well without heavy tuning

So it gives reliable yield predictions, which is essential before explaining anything.

STAGE 2

Agentic Feedback Loop (Rounds 1–10)

Each round is a 3-step cycle that iteratively improves the recommendations.

1 Gap Analysis (Prompt 2)



“What analyses are needed?”

LLM reviews previous output, identifies what’s missing or unclear, and decides what to analyze.



LLM generates Python code

(e.g., correlations, PCA, clustering, interaction analysis)

2 Code Execution



Code executes on the real dataset.
Outputs (tables, plots, statistics) are compiled into a PDF report.



PDF of Figures / Stats

Evidence generated from the data.

3 Synthesis (Prompt 3)



“Rewrite recommendations with new evidence.”

LLM receives the new PDF + original SHAP plot + full conversation history.



Rewritten Recommendations

Incorporates new evidence with greater depth and clarity.

Enhanced Recommendations (Rounds 1–10)

Progressively more refined and evidence-based.

3. Quality Evaluation

Blind evaluation of recommendations



Crop Scientists (n = 12)

Expert evaluation of clarity, actionability, depth, and usefulness.

+



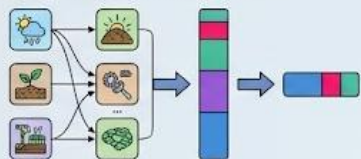
LLMs as Evaluators (n = 14)

Multiple advanced LLMs evaluate using the same criteria.

4. Explainability Techniques

SHAP (SHapley Additive exPlanations) — TreeExplainer variant

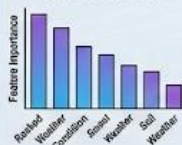
1. Shapley Value Calculation



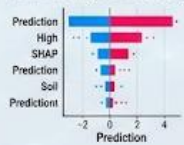
Assigning each feature a Shapley value — the average marginal contribution of that feature across all possible feature subsets

2. Dual Interpretability

Global Interpretability
(which features matter most overall)



Local Interpretability
(how features affect individual predictions)



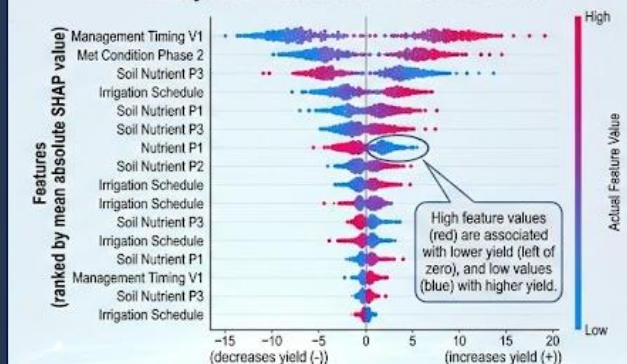
SHAP is the core XAI backbone.

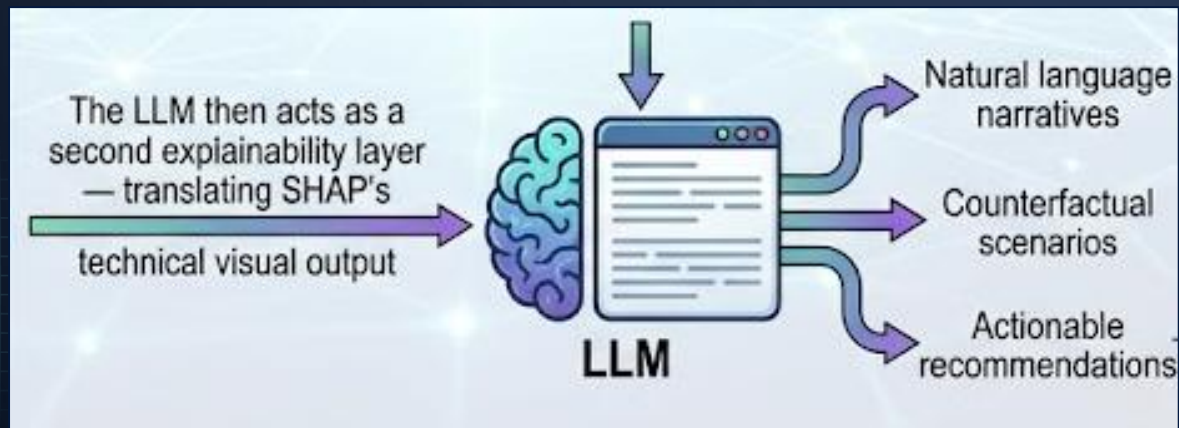
Why SHAP?

- Provides both global and local explanations
 - Consistent feature attribution
 - Works effectively with Random Forest models
 - Widely adopted in XAI research*
- Produces visual outputs interpretable by both humans and LLMs

SHAP serves as the starting point of the Agentic XAI workflow by providing the evidence used for iterative refinement.

Primary Visualization: SHAP Beeswarm Plot





5. Dataset and Evaluation Metrics

Property	Detail
Crop	Rice (4 varieties: Fukunoka, Gohyakugawa, Sakurafukuhime, Tennotsubu)
Location	Tomioka-machi, Fukushima, Japan (37.34°N, 141.00°E)
Period	2021–2023 (3 years)
Scale	26 fields, ~9 hectares
Observations	n = 66 field-year records
Yield measurement	UAV imagery + deep learning (Xception model) + IDW interpolation

37 explanatory variables across three categories:

- 1. Soil properties (7 variables):** organic matter, total carbon, total nitrogen, EC, sand, C:N ratio, available phosphorus
- 2. Agricultural management (10 variables):** variety, days to heading, planting system, organic farming, green manure, manure, soil amendments, herbicide, pesticide, years of production
- 3. Meteorological conditions (20 variables):** temperature (avg/min/max), precipitation, solar radiation ,each across 4 growth stages

ML Model

- Random Forest** with LOO-CV;
- optimal config**: 100 estimators, max depth = 10
- Cross-validation** $R^2 = 0.749$; training $R^2 = 0.962$ (some overfitting noted, but adequate for SHAP grounding)



Metric

Clarity
Conciseness
Contextual
Relevance
Cost Consideration
Crop Science
Credibility
Practicality
Specificity

What it measures

Structure, readability, logical flow
Avoids verbosity and redundancy
Fits the specific dataset, location, and scenario
Addresses economic factors and ROI
Agronomic soundness and scientific accuracy
Actionability with typical farm resources
Precision : specific numbers, varieties, methods

6- Reproducibility Assessment

Component	Status
Paper	Available
Notebook	Available
Prompts	Available
Figures	Available
Recommendations	Available
Dataset	Available Upon Request

The study provides most materials required for reproduction.

The original dataset is available upon request from the authors.

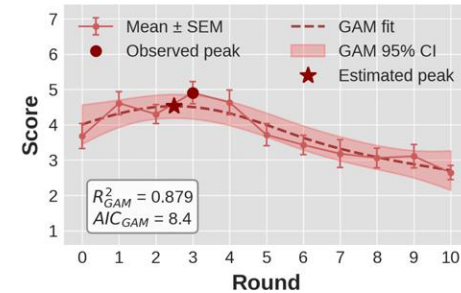
- The authors evaluated recommendations from each iteration round using assessments from 12 crop scientists and 14 LLM evaluators.
- They used ANOVA to determine whether recommendation quality changed significantly across rounds.
- Then, they fitted a Generalized Additive Model to test their main hypothesis that explanation quality follows an inverted U-shaped pattern, improving initially and then declining when explanations become overly detailed.
- Finally, they compared the GAM against a simple linear model using AIC and R^2 to ensure that any nonlinear pattern was genuinely supported by the data.

The authors used a Generalized Additive Model (GAM) to determine whether a metric truly had: Improve → Peak → Decline instead of just random fluctuations.

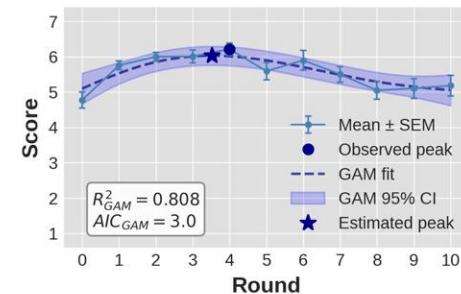
Why GAM?

The authors hypothesized that explanation quality would not change linearly across refinement rounds. A Generalized Additive Model (GAM) was used to detect potential non-linear patterns and verify whether recommendation quality follows an inverted U-shaped trajectory (Improve → Peak → Decline).

(a) Crop Scientists (n = 12)



(b) LLMs (n = 14)



7. Experimental Results

Main Finding : Inverted U-Shaped Quality Trajectory

Both evaluator groups independently confirmed the same pattern: quality peaks at Rounds 3–4, then degrades sharply.

Group	Round 0 Score	Peak Score	Peak Round	Round 10 Score	% Improvement to Peak
Crop Scientists	3.68	4.91	Round 3	2.64	+33%
LLMs	4.78	6.21	Round 4	5.18	+30%

GAM (Generalized Additive Model) analyses confirmed the inverted U-shape statistically in both groups ($\Delta AIC = 9.25$ for crop scientists, 13.42 for LLMs both well above the threshold of 2).

Metric-Specific Patterns :Three Distinct Behaviors:

What happens to explanation quality as the agent keeps refining explanations from Round 0 to Round 10?

Aggregated behavior across BOTH evaluators (human + LLM)

Group 1 : Inverted U-shape (5 metrics): Specificity, Clarity, Practicality, Contextual Relevance, Crop Science Credibility

- Peaked at Rounds 2–4 then declined
- These metrics are most sensitive to over-refinement

This directly supports the paper's hypothesis: There is an optimal explanation depth. More explanation is not always better.

Group 2 :Monotonic Increase (Cost Consideration):

- Kept improving through Round 9–10
- Critical finding: the dataset had *no economic variables*, yet the LLM kept generating economic analyses , a clear case of ungrounded abstraction (high variance)

As the AI performed more analyses, it became better at understanding:

- farm constraints
- management decisions
- trade-offs

So cost-awareness kept improving throughout the 10 rounds. There was no decline

Group 3 :Monotonic Decrease (Conciseness):

- Already declining from Round 0 onward
- Captures progressive verbosity, the most direct evidence of the variance side of the trade-off

Conciseness means: Is the explanation short and to the point? What happened?

As rounds increased: The AI kept adding: correlations, PCA findings, clustering results, additional evidence. Therefore explanations became longer and longer

For the five main metrics:
GAM detected a real inverted U-shape.

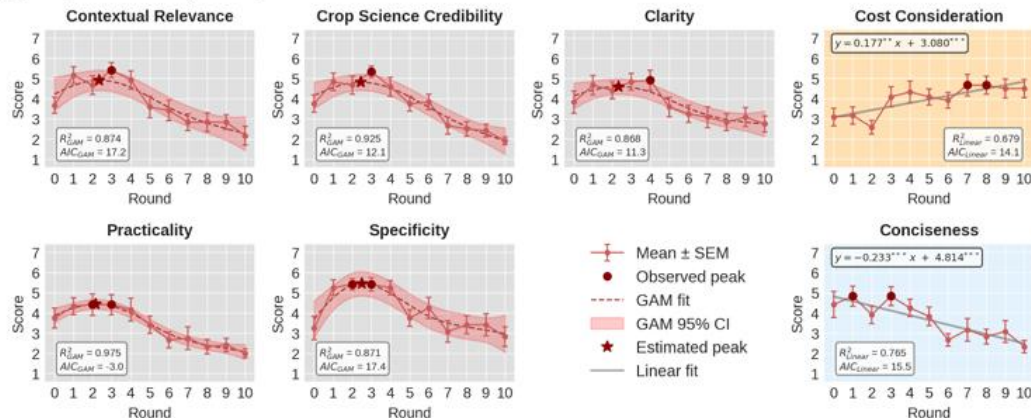
For Cost Consideration:
No inverted U-shape found. Only a positive trend.

For Conciseness: No inverted U-shape found.
Only a negative trend.

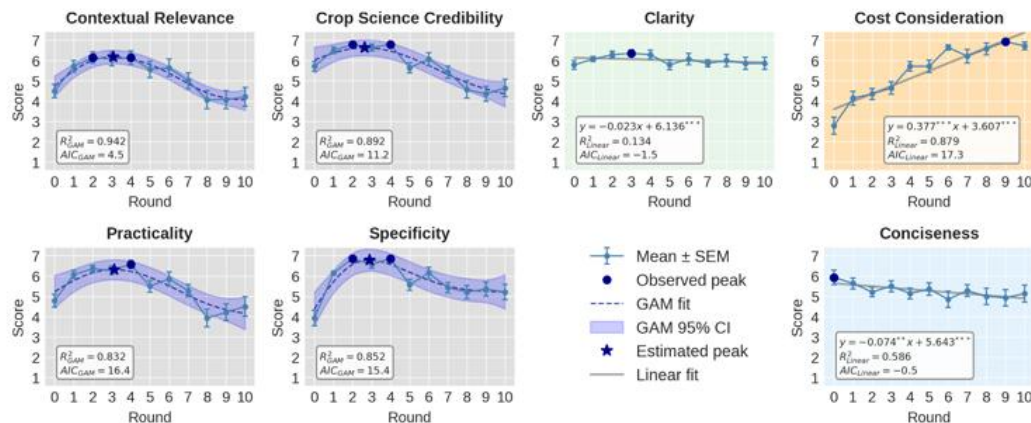
Interpretation

1. Most quality metrics improved initially and then declined due to over-refinement, supporting the bias-variance trade-off hypothesis.
2. Cost Consideration continued to increase because the LLM progressively introduced economic reasoning, even when economic variables were absent from the dataset.
3. Conciseness continuously decreased because each refinement round added more analyses, evidence, and discussion, resulting in increasingly verbose explanations.

(a) Crop Scientists (n = 12)



(b) LLMs (n = 14)



Analytical Complexity Progression

Round 0: 1 figure (beeswarm), basic SHAP interpretation

Round 3: 36 cumulative figures, PCA, response curves, error bars

Round 7: 71 figures, K-S tests, polynomial regression, critical thresholds

Round 10: 93 cumulative figures, beta distribution simulations, exit criteria but quality is at its worst

This shows the agent became analytically more sophisticated while becoming practically less useful a textbook variance overfit

As the agent performed more analyses, explanation complexity increased from 1 figure in Round 0 to 93 figures in Round 10. While early rounds improved recommendation quality, excessive analytical complexity eventually reduced clarity and usability, supporting the paper's hypothesis that there is an optimal level of explanation depth

This table is using the bias–variance trade-off from machine learning as an analogy for how “analysis quality” changes across different phases (or rounds).

Phase	Rounds	Problem	Analogy
Early	0–2	Lacks depth, too general	High bias
Optimal	3–4	Balanced depth and clarity	Sweet spot
Late	5–10	Verbose, ungrounded, abstract	High variance

- Too simple → high bias → underfitting
- Too complex → high variance → overfitting
- Balanced → best generalization

Key Design Principles Derived:

The authors conclude with four evidence-based principles for trustworthy agentic XAI systems:

1. **Strategic early stopping** (3–4 rounds optimal here)
2. **Hybrid human-LLM evaluation for scalable yet credible assessment** (they propose combining both: Humans handle final judgment / validation. LLMs handle large-scale screening and intermediate evaluation)
3. **Observability protocols:** archive intermediate code and visualizations for expert verification (Agentic systems go through multiple internal steps (code, reasoning, visual outputs). The authors say you should store and track intermediate artifacts, such as: generated code intermediate results visualizations (e.g., plots, heatmaps))
4. Recognize that metrics cannot all be simultaneously optimized through iteration alone (So more iteration does NOT guarantee better overall performance)

Trustworthy agentic XAI is not about making systems more complex or iterative, it's about knowing when to stop, how to evaluate, and how to preserve transparency.

Thank you !

Do you have any questions?

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